Algebraic Topology: a beginner's course

This lecture:

AlgTop9: Applications of Euler's formula and graphs


Lecture 9: Applications of Euler's formula
and graphs

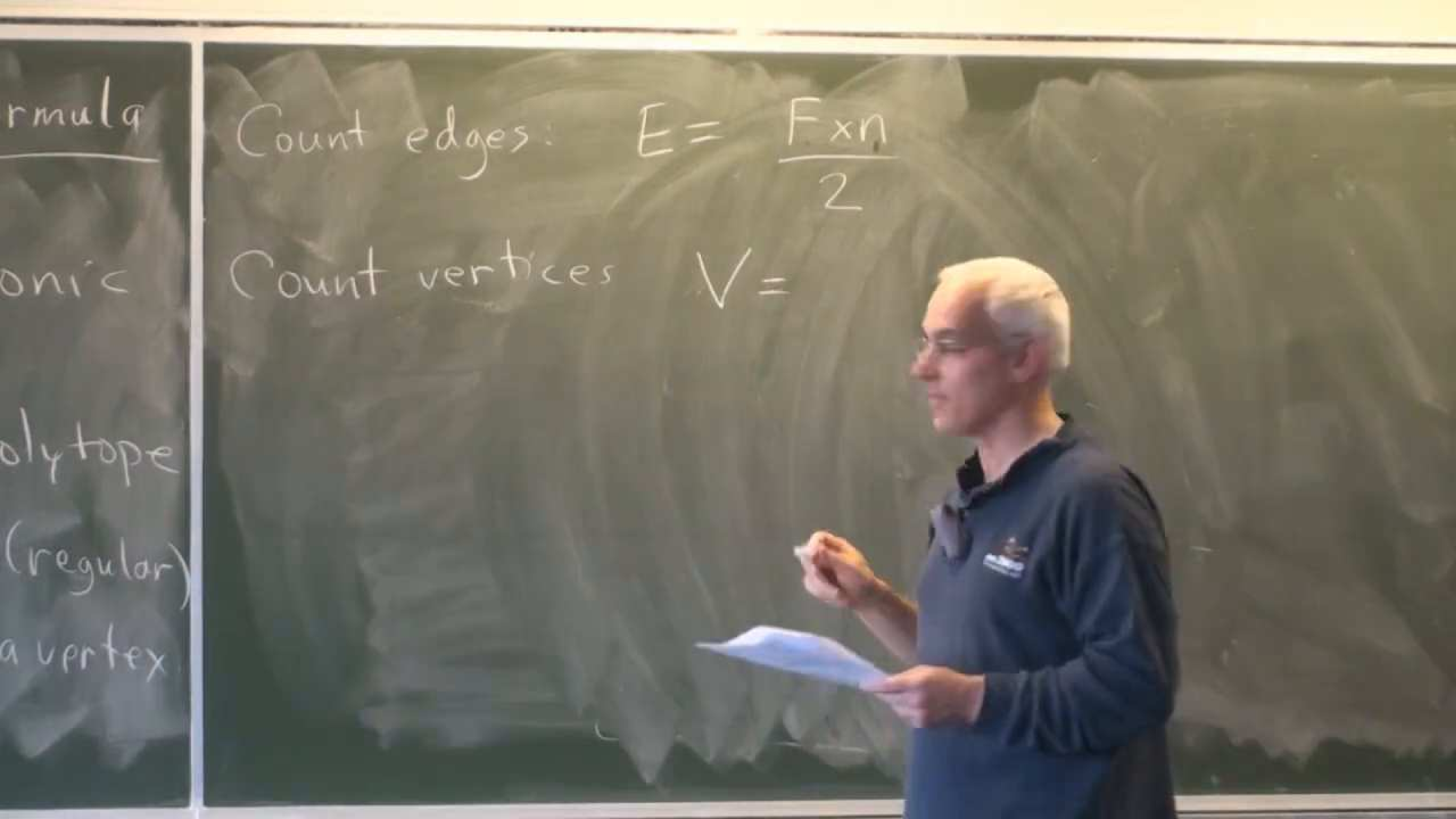
Lecture 9: Applications of Euler's formula and graphs

Thm. There are at most 5 Platonic solids.

Proof. Assume have a regular polytope

$n =$ # edges on each face (regular)

$m =$ # edges that meet at a vertex


rmula
onic
olytope
(regular)
a vertex
Count edges: $E = \frac{F \times n}{2}$
Count vertices $V =$

$$E = \frac{F \times n}{2}$$

es $$V = \frac{F \times n}{m}$$

$$= 2$$

$$+ F = 2$$

so $F\left(\frac{n}{m} - \frac{n}{2} + 1\right) = 2$

$$F = \frac{4m}{2n - mn + 2m}$$

must have $2n - mn + 2m > 0$

or $\boxed{2(n+m) > nm}$

$2n > m(n-2)$ $\quad m, n \geq 3$

$\frac{2n}{n-2} > m \geq 3 \Rightarrow 2n > 3n - 6$ $\quad \boxed{6 > n}$

so also $\boxed{6 > m}$

possibilities are

$(n,m) =$ $(3,3)$ T

$(3,4)$ O

$(3,5)$

$(4,$ [illegible]$)$

$(5,3)$

Count edges: $E = \dfrac{F \times n}{2}$

Count vertices $V = \dfrac{F \times n}{m}$

$V - E + F = 2$

$\dfrac{Fn}{m} - \dfrac{Fn}{2} + F = 2$

so $F\left(\frac{n}{m} - \frac{n}{2} + 1\right) = 2$

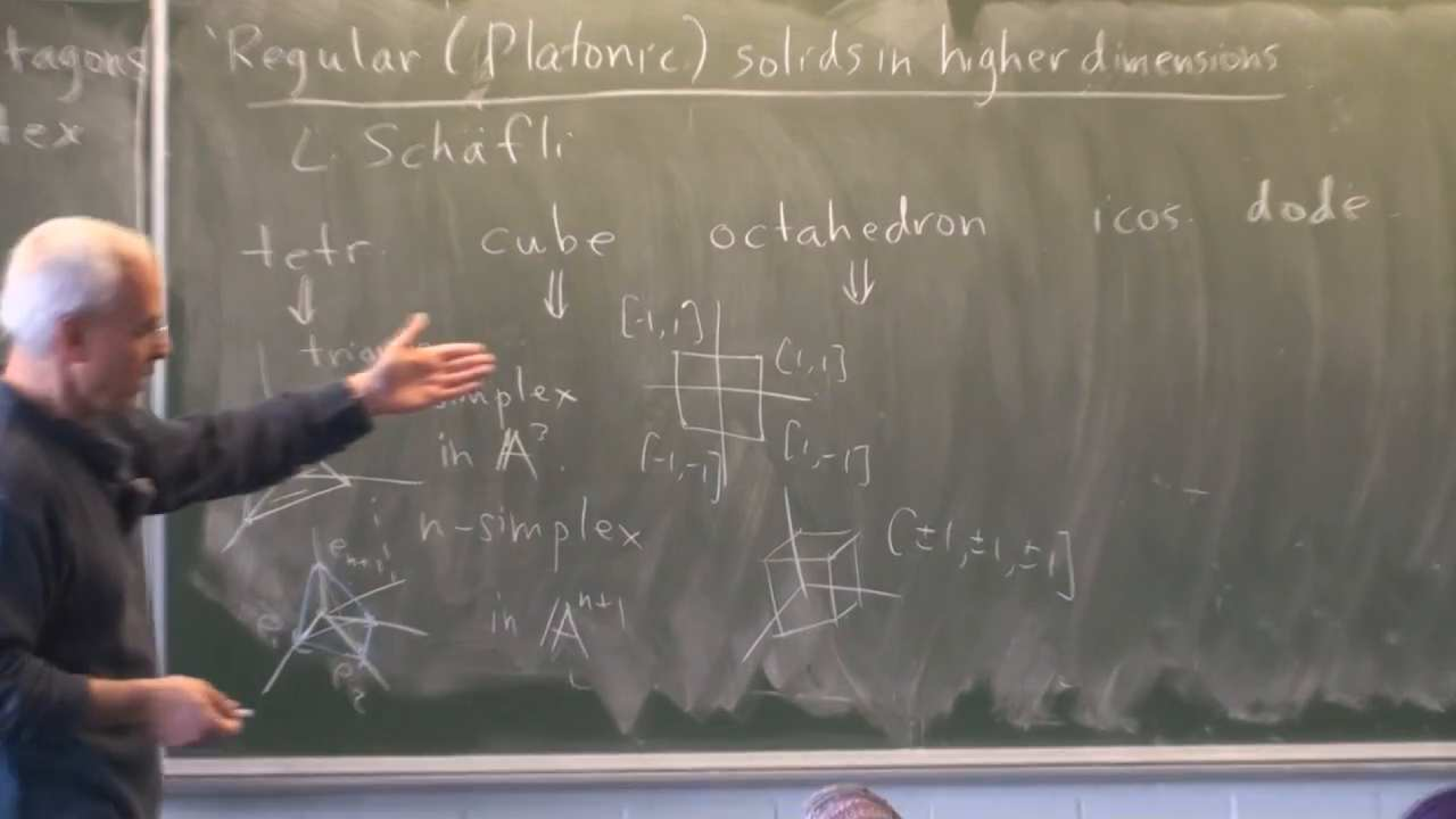
$F = \dfrac{4m}{2n - mn + 2m}$

must have $2n - mn +$

or $\boxed{2(n+m) > nm}$

$2n > m(n-2)$ $\quad m, n \geq$

$\dfrac{2n}{n-2} > m \geq 3 \implies 2n$

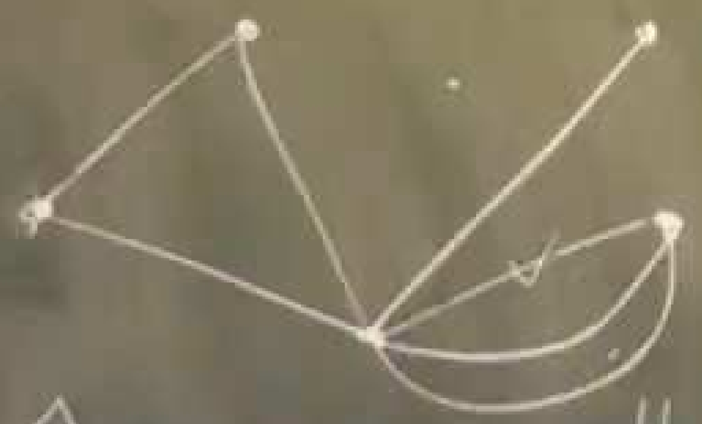

Regular (Platonic) solids in higher dimensions
L. Schäfli
tetr.
cube
octahedron
icos.
dode
simplex
in $\mathbb{A}^3$
n-simplex
in $\mathbb{A}^{n+1}$
e_{n+1}
e_1
e_2
[-1,1]
[1,1]
[-1,-1]
[1,-1]
[±1,±1,±1]

Graphs

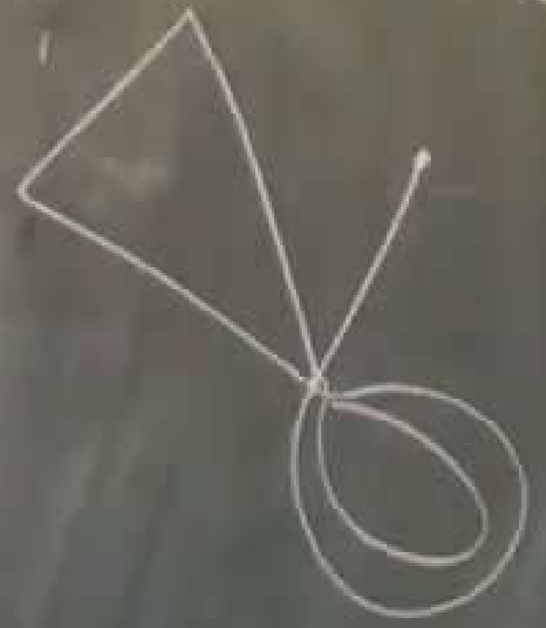
ell

120

Graphs



parallel edges



simple graph: no loops
no parallel edges

assume connected

$V = 5$

$E = 8$

$F = 5$ (outside included)

$\boxed{V - E + F = 2}$

Thm Fo

V-E+F

Proof: T

If it is a

Then E→

If it is joi

shrink it so v

E→E-1 ,